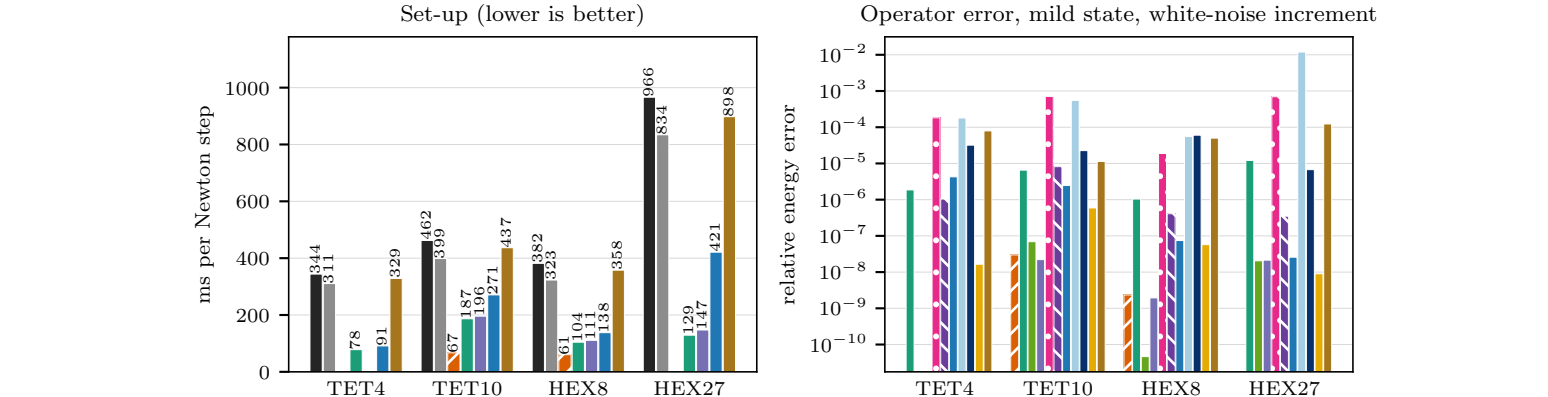

Set-up (lower is better)
ms per Newton step
0
200
400
600
800
1000
344
311
78
91
329
462
399
67
187
196
271
437
382
323
61
104
111
138
358
966
834
129
147
421
898
TET4
TET10
HEX8
HEX27
Operator error, mild state, white-noise increment
relative energy error
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
10^{-10}
TET4
TET10
HEX8
HEX27
BSR, fp64
BSR, fp32
projected (approx.)
points/moments, fp16
points/moments, int8 mean+dev.
element matrix, fp16 diff.
quantized BSR, int8
points/moments, fp16 mean+dev.
points/moments, int8 per element
element matrix, fp16 full
element matrix, int8 diff.
element matrix, int8 diff., pair scale
quantized BSR, fp16